Dynamic MoE-DU-GAN with a Noise Expert for Medical Image Denoising

At Iter=100k on a combined CT+MRI+Ultrasound training set, CBAM gives the best PSNR/RMSE and Hybrid gives the best SSIM. I also implemented a MoE version with a Noise Expert and a gating router. The MoE model is trained with a three-phase schedule and reaches Iter=200k on the combined set.

作者: 吳定栩 B11132015　　指導教授: 鮑興國　國立臺灣科技大學 資訊工程系　　*2025/12/24*

BACKGROUND

Low-dose CT helps reduce radiation, but it also brings more noise and artifacts. The goal is to remove noise while keeping small anatomical details.

BASELINE (DU-GAN)

DU-GAN uses a generator plus two discriminators:

- an image-domain discriminator to push outputs toward clean CT appearance
- a gradient-domain discriminator to keep edges sharp

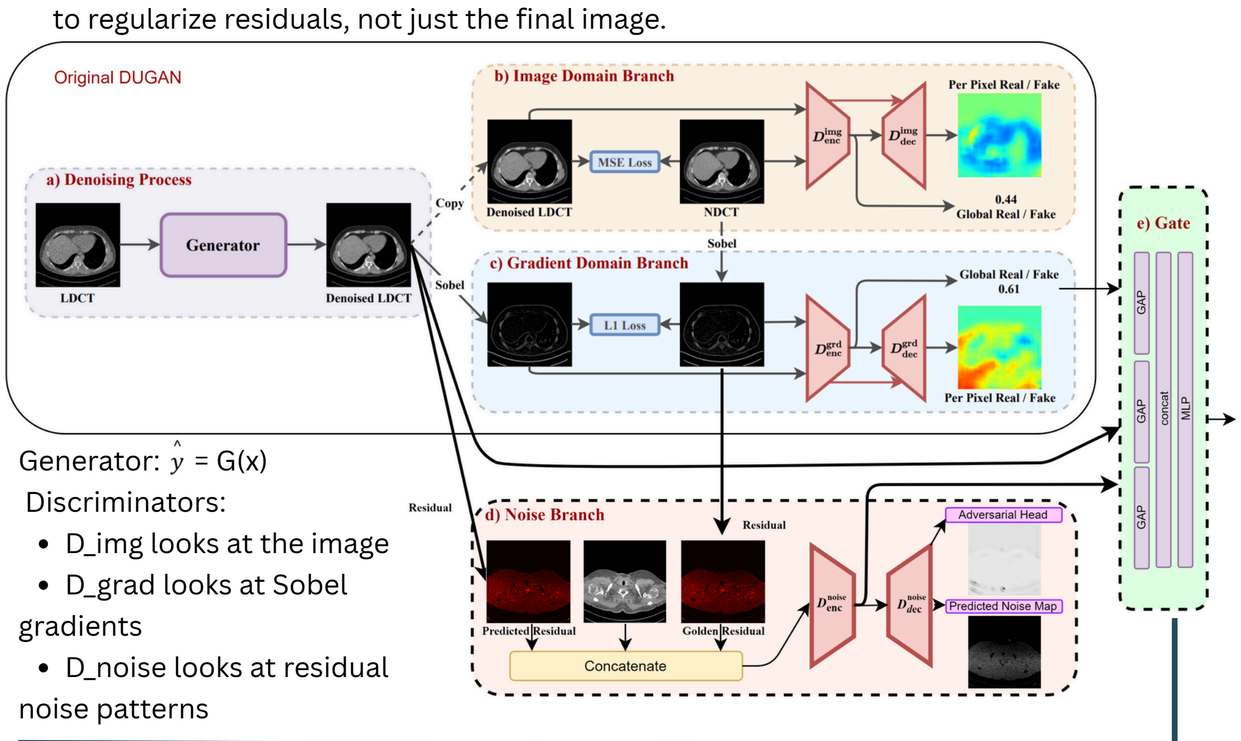
A common issue is that the loss weights are fixed, so the same weighting is used for every case, even when noise level or modality changes.

WHAT I DID

- Discriminator wrapper upgrades (CBAM, Hybrid variant, Frequency attention), replacing original image-discriminator
- a Dynamic Mixture-of-Experts setup: add a Noise Expert and a gating network to reweight the adversarial losses during training, aiming higher noise attention on multi-modal cases (e.g., MRI)

METHOD OVERVIEW

- Given input x, the generator outputs G(x). DU-GAN uses image and gradient discriminators to enforce realism and protect edges. Given x, $\hat{y}$ = G(x). I add a noise expert to regularize residuals, not just the final image.



Generator: $\hat{y}$ = G(x)

Discriminators:

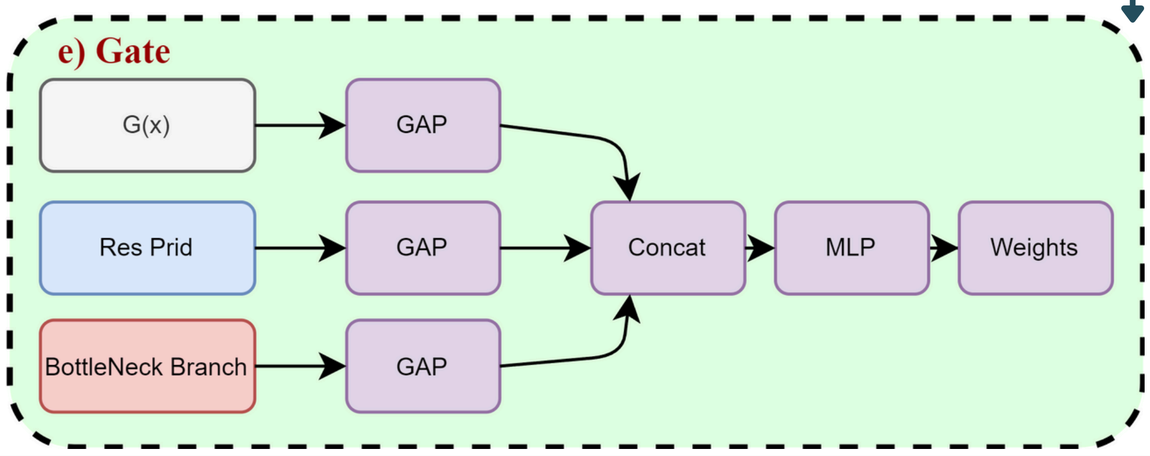
- D_img looks at the image
- D_grad looks at Sobel gradients
- D_noise looks at residual noise patterns

NOISE EXPERT (d)

Instead of only judging "does the final image look real, " the noise expert checks the residual: r_pred = |x − ŷ|, If anatomy gets removed, r_pred will contain structure-like patterns. The noise expert supervises residual realism and predicts a noise-strength map, and its bottleneck features are also fed to the gate.

It produces:

- a per-pixel real/fake map for adversarial supervision on residual realism
- a predicted noise map with a regression loss (trained against a proxy residual map)



GATING NETWORK (e)

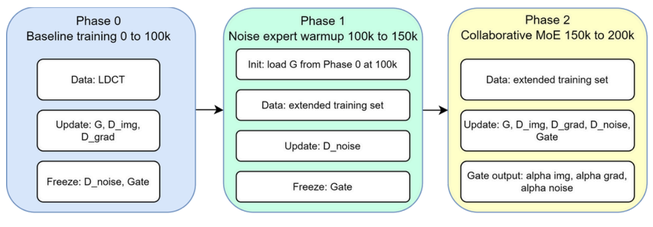
The gate outputs mixture weights: $\boldsymbol{\alpha} = [\alpha_{\text{img}}, \alpha_{\text{grad}}, \alpha_{\text{noise}}], \sum\alpha = 1$

It takes:

- the denoised output ŷ
- the predicted residual |x − ŷ|
- bottleneck features from the noise expert

Then it reweights the adversarial losses: $\mathcal{L}_{adv} = \alpha_{img}\mathcal{L}_{adv,img} + \alpha_{grad}\mathcal{L}_{adv,grad} + \alpha_{noise}\mathcal{L}_{adv,noise}$

TRAINING (THREE PHASES)



Phase 0 (0–100k): train baseline DU-GAN on CT

Phase 1 (100k–150k): freeze G, train the Noise Expert on the combined set

Phase 2 (150k–200k): train G + Gate + all discriminators on the combined set

DATA

CT (AAPM-Mayo): train 300,000 patches (64×64), test 6,590 images (512×512)

MRI appendix: 9,000 samples (resized to 64×64)

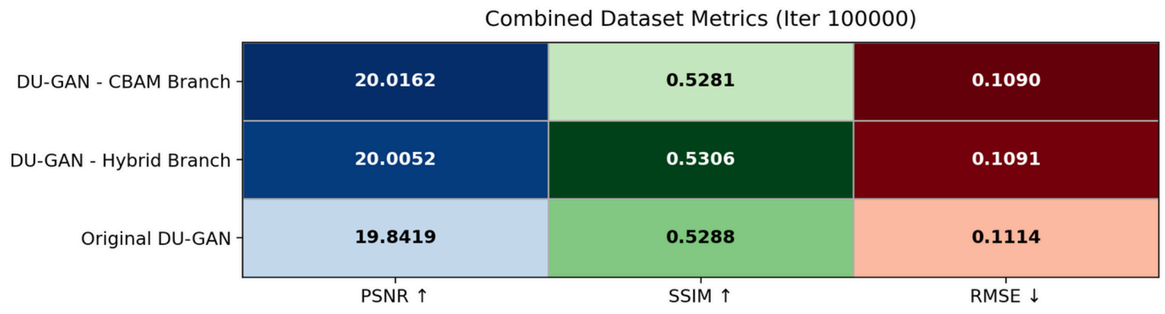
Ultrasound appendix (BUSI): 780 samples (resized to 64×64)

Combined training set size: 309,780 samples

METRICS

PSNR (higher), SSIM (higher), RMSE (lower)

RESULTS ON COMBINED DATASET



How to read this table

This heatmap is the main head-to-head comparison on the combined training distribution at the same checkpoint (Iter=100k). Each row is one model variant and each column is one metric. PSNR/SSIM should be higher and RMSE should be lower.

CBAM is best on PSNR/RMSE (PSNR 20.0162, RMSE 0.1090). Hybrid is best on SSIM (0.5306). The gains are small, but they are consistent under the same 100k setting.

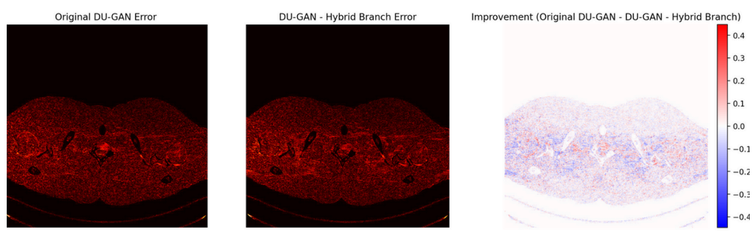
I extend the training mix by appending MRI and BUSI to the LDCT patch set. For MoE, I start from the CT-only checkpoint at 100k (Phase 0), then run Phase 1 (train the Noise Expert with G frozen) and Phase 2 (train G + Gate + all discriminators). Rows marked (Extended) indicate continued training from the LDCT-only pretrained checkpoint on the multi-modal training set.

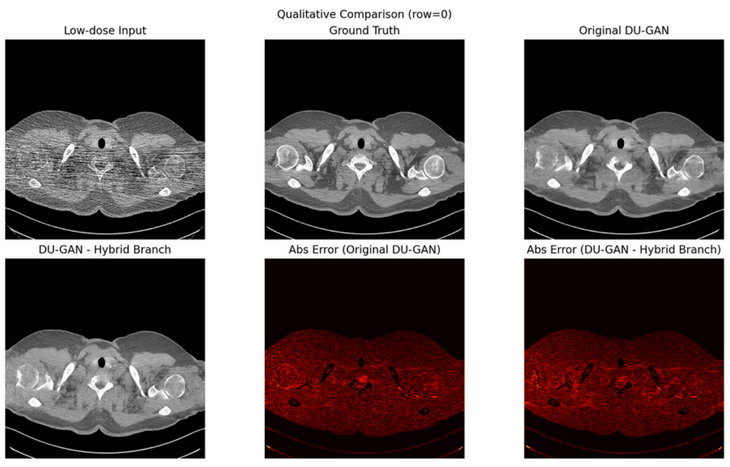
RESULTS ON CT-ONLY DATASET (100k iters)

Model / Run	Iter	PSNR	SSIM	RMSE
DU-GAN Original	100k	21.9160	0.7464	0.0835
DU-GAN Original(Extended)	150k	21.6811	0.7443	0.0859
DU-GAN + CBAM	100k	22.0711	0.7461	0.0820
DU-GAN + Hybrid	100k	22.1491	0.7476	0.0813
DU-GAN + Hybrid (Extended)	150k	21.6193	0.7406	0.0866
DU-GAN + Frequency Attn	50k	22.1367	0.7465	0.0814
DU-GAN + Frequency Attn(Extended)	150k	21.9229	0.7419	0.0832
DU-GAN + CBAM (Extended)	150k	22.0871	0.7474	0.0820
DU-GAN MoE	200k	23.3498	0.7322	0.0707

Notes on the CT-only table

This table aggregates multiple runs at different iterations. Only rows at the same iteration are a fair ranking; the cleanest comparison is the 100k block (Original / CBAM / Hybrid). Other rows (Wavelet + CA, FreqAtt, Extended CBAM, MoE) are kept to show explored directions and training-time trends, not as a strict leaderboard.





Qualitative panel

The qualitative panel answers: where is noise reduced, and are edges preserved? The residual/improvement maps help reveal small changes without heavy zooming.

When reading, check (1) flat regions for leftover noise and (2) boundaries for over-smoothing.

CONCLUSION

I propose a dynamic MoE extension of DU-GAN by adding a noise expert and a gating network to adaptively balance image, gradient, and noise based supervision. A three-phase training strategy stabilizes optimization and enables effective collaboration between experts. On the combined-set sanity check at Iter=100k, CBAM achieves the best PSNR/RMSE and Hybrid achieves the best SSIM, showing consistent gains under domain shift compared to the original DU-GAN. On the CT-only test set, MoE achieves the best scores at Iteration = 200k, a multi-modal benchmark is left for future work.

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